NUCLEAR FUEL CYCLE

Transport of Radioactive Material

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- Radioactive material accounts for a very small proportion of all dangerous material shipped each year – just 1% in the USA, the world's largest producer of nuclear power.
- Globally, about 15 million packages of radioactive material are transported each year on public roads, railways, and ships.
- Radioactive material is not unique to the nuclear fuel cycle. The significant majority – about 95% – of radioactive consignments are not related to nuclear power.
- Transport is, however, an integral part of the nuclear fuel cycle; most countries that mine uranium do not produce nuclear power.
- Though transport is a very minor cost in the nuclear fuel cycle, lack of harmonization and over-regulation in authorization creates problems for transport of all radioactive materials between countries.

Radioactive material constitutes only a very small proportion of all dangerous goods. In the USA – a country that operates over 20% of the world's civil nuclear power reactors – flammable, explosive, corrosive, or poisonous materials account for 99% of the dangerous material shipped each year.[a] According to the US Department of

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Transportation, less than 0.5% of the total cost related to the shipping of dangerous materials each year in the USA is attributable to radioactive substances.[b]

International regulations for the transport of radioactive material have been published by the International Atomic Energy Agency (IAEA) since 1961. The IAEA has regularly issued revisions to the transport regulations in order to keep them up-to-date. The latest edition of its Regulations for the Safe Transport of Radioactive Material was released in 2018. The IAEA regulations have been widely adopted into national policies, as well as by the International Civil Aviation Organization (ICAO), the International Maritime Organization (IMO), and regional transport organizations.

The objective of the regulations is to protect people and the environment from the effects of radiation during the transport of radioactive material, both routinely and when transport accidents occur. The fundamental principle is that the protection comes from the design of the package, regardless of how the material is transported. More specifically, protection is achieved by:

- Containment of radioactive contents.
- Control of external radiation levels.
- Prevention of criticality.
- Prevention of damage caused by heat.

Radioactive material (defined as Class 7 (see Notes) material in the UN Model Regulations on Dangerous Goods) is not unique to the nuclear fuel cycle. Radioactive substances are used extensively in medicine, agriculture, research, manufacturing, non-destructive testing, and minerals exploration (for further information, see information page on The Many Uses of Nuclear Technology), and it is estimated that just 5% of radioactive material shipped globally each year relates to nuclear power production.[a] Regulatory control of shipments of radioactive material is independent of the material's intended application.

About 15 million packages of radioactive material are transported around the world each year. Since 1961, when the IAEA's safe transport regulations were first issued, it is likely that over one billion consignments have been safely completed.[a] Of the approximately three million packages containing radioactive material that are shipped each year in the USA, only about 250,000 contain waste from US nuclear power plants, and just 25 to 100

contain used fuel.[c] The US Department of Transportation estimates that the average distance per shipment of radioactive material is about 55 kilometres, well below the 185 kilometre average across all hazardous materials.[b]

For the generation of a given quantity of electricity, the amount of nuclear fuel required is very much smaller than the amount of any other fuel. Therefore, the conventional risks and environmental impacts associated with fuel transport are greatly reduced with nuclear power (for further information, see information page on Nuclear Energy and Sustainable Development). Nevertheless, whilst volumes of fuel required are modest, used fuel emits very high levels of heat and radioactivity. As such, consignments are transported in robust 125-tonne 'Type B' casks, each containing up to about 20 tonnes of used fuel (see below). The Nuclear Regulatory Commission notes: “Over the last 40 years, thousands of shipments of commercially generated spent nuclear fuel have been made throughout the USA without causing any radiological releases to the environment or harm to the public.”[d] Most of these shipments are between different power plants owned by the same utility, so as to share storage space for spent fuel.

Materials requiring transport

Transport is an integral part of the nuclear fuel cycle. There are about 440 operable nuclear power reactors in 32 countries, but uranium mining occurs in only about 30, with most production from countries without nuclear power. Furthermore, in the course of over 60 years of the nuclear power industry, a number of centralized facilities have been developed in various locations around the world to provide fuel cycle services.

Most material used in nuclear fuel is transported several times during its progress through the fuel cycle. Transport is frequently international and often over large distances, but is a very minor cost in the overall fuel cycle. Much of the material moved is similar to that from other industrial activities. However, fresh nuclear fuel, which is mildly radioactive, and some wastes that are significantly more so, are the focus of significant attention. Any substantial quantity of radioactive material is generally transported by specialized companies. (For further information on nuclear waste, see information page on Radioactive Waste Management).

The term 'transport' is used in this document to refer only to the movement of material between facilities (*i.e.* through areas outside

such facilities' boundaries). Most consignments of nuclear fuel material occur between different stages of the cycle, but occasionally material may be transported between similar facilities. When the stages are directly linked (such as mining and milling) the facilities for the different stages are usually on the same site, and no transport is required.

With very few exceptions, nuclear fuel cycle materials are transported in solid form. The following table shows the principal nuclear material transport activities.

Stages of nuclear transport

From:	To:	Material	Notes
Mining	Milling	Ore	Rare: usually on the same site
Milling	Conversion	Uranium oxide concentrate ('yellowcake')	Usually 200-litre drums holding 400 kg, in standard six-metre transport containers
Conversion	Enrichment	Natural uranium hexafluoride (UF_6)	Special UF_6 containers, Type 48Y
Enrichment	Fuel fabrication	Enriched UF_6	Special UF_6 containers, Type 30B
Fuel fabrication	Power generation	Fresh (unused) fuel	Type A casks unless MOX fuel (Type B)
Power generation	Used fuel storage	Used fuel	After onsite storage, large Type B casks
Used fuel storage	Disposal*	Used fuel	Large Type B casks
Used fuel storage	Reprocessing	Used fuel	Large Type B casks
Reprocessing	Conversion	Uranium oxide	Called reprocessed uranium (RepU)
Reprocessing	Fuel fabrication	Plutonium oxide	
Reprocessing	Disposal*	Fission products	Vitrified (incorporated into glass)
All facilities	Storage/disposal	Waste materials	Sometimes on the same site

*Not yet taking place

Although some waste disposal facilities are located adjacent to the facilities that they serve, using one disposal site to manage the wastes from several facilities usually reduces environmental impacts and associated costs. When shared disposal is used, transport from individual facilities to the disposal site will be required. (For further information, see information paper on Storage and Disposal of Radioactive Wastes.)

Uranium oxide concentrate ('yellowcake')

Uranium oxide concentrate, sometimes called yellowcake, is transported from mines to conversion plants. Transport takes place in 200-litre drums, each holding about 400 kg U_3O_8, packed into normal six-metre shipping containers. No radiation protection is required beyond having the steel drums clean and within the shipping container.

The importance of safe and secure yellowcake transport is evidenced by the fact that 80% of uranium is mined in just five countries, only one of which (Canada) uses uranium for nuclear power.

Every year around 50 shipments of 500 containers of Australian uranium are safely transported by road and rail to ports in Adelaide and Darwin where they are shipped, with no recorded incident affecting public health. Australia is the world's third largest miner of uranium.[e]

Natural and enriched UF_6

To and from enrichment plants, uranium is in the form of uranium hexafluoride (UF_6), which has low levels of radioactivity, but significant chemical toxicity. Natural uranium as hexafluoride is usually shipped to enrichment plants in Type 48Y cylinders, each 122 cm diameter and holding about 12.5 tonnes of UF_6 (8.4 tU). These cylinders are then used for long-term storage of depleted uranium as hexafluoride, typically at the enrichment site. Enriched uranium is shipped to fuel fabricators in smaller Type 30B cylinders, each with a 76 cm diameter and holding 2.27 t UF_6 (1.54 tU). High-assay LEU (HALEU) above 5% enriched will require new cylinders.

Fresh fuel

Uranium fuel assemblies are manufactured at fuel fabrication plants. The fuel assemblies are made up of ceramic pellets formed from pressed U_3O_8 that has been sintered at a high temperature (over 1400°C). The pellets are aligned within long, hollow metal rods, which in turn are arranged in the fuel assemblies, ready for introduction into the reactor.

Different types of reactor require different types of fuel assembly, so when the fuel assemblies are transported from the fuel fabrication facility to the nuclear power reactor, the contents of the shipment will vary depending on the type of reactor receiving it.

In Western Europe, Asia, and the USA, the most common means of transporting uranium fuel assemblies is by truck. A typical truckload supplying a light water reactor contains six tonnes of fuel. In Russia and Eastern Europe rail transport is most often used. Intercontinental transport is mostly by sea, though occasionally by air.

The annual operation of a 1000 MWe light water reactor requires an average fuel load of 27 tonnes of uranium dioxide, containing 24 tonnes of enriched uranium. The required annual fuel load can be transported in 4 or 5 trucks. The fuel assemblies are transported in packages specially constructed to protect them from damage during transport. Uranium fuel assemblies have a low radioactivity level and radiation shielding is not necessary.

Fuel assemblies contain fissile material and criticality is prevented by the design of the package, and the maximum number of packages carried in one shipment.

Used fuel

Used fuel from a nuclear power reactor contains 96% uranium, 3% fission products, and 1% plutonium as well as a small amount of other transuranics.

Used fuel emits high levels of both radiation and heat, and so is stored in water pools adjacent to the reactor to allow the initial heat and radiation levels to decrease. Typically, used fuel is stored onsite for at least five months before it can be transported, although it may be stored there long-term. From the reactor site, used fuel is transported by road, rail, or sea to either an interim storage site or a reprocessing plant.

Used fuel assemblies are shipped in Type B casks which are shielded with steel, or a combination of steel and lead. Since 1971

there have been at least 25,000 cargoes of used fuel transported, covering many millions of kilometres on both land and sea.[f] In total some 300 sea voyages have been made in purpose built ships carrying used nuclear fuel over a cumulative distance of more than 8 million kilometres. The major company involved has completed about 200 sea voyages over 40 years, transporting more than 4000 casks, each of about 100 tonnes.[g]

Shipments of spent fuel from Japan to Europe for reprocessing used 94-tonne Type B casks, each holding a number of fuel assemblies. More than 160 of these shipments took place from 1969 to the 1990s, involving more than 4000 casks, and moving several thousand tonnes of highly radioactive used fuel to the UK (4200 tonnes) and to France (2940 tonnes). (For further information see information paper on Japanese Waste and MOX Shipments From Europe.)[h]

In Sweden, more than 80 large transport casks are transported annually to the CLAB central interim waste storage facility. Each 80-tonne cask has steel walls 30 cm thick and holds 17 BWR or 7 PWR fuel assemblies. The used fuel is shipped to CLAB after it has been stored for about a year at the reactor, during which time heat and radioactivity diminish considerably. Over 7000 tonnes of used fuel has been transported to CLAB, much of it around the coast by ship.[i]

In the USA the Nuclear Regulatory Commission noted about 2013: "Over the last 40 years, thousands of shipments of commercially generated spent nuclear fuel have been made throughout the United States without causing any radiological releases to the environment or harm to the public." Most of these shipments are between different power plants owned by the same utility, so as to share storage space for used fuel.

Canada's Nuclear Waste Management Organization published a report in 2015 estimating spent nuclear fuel consignments worldwide:

- Canada: five per year by road.
- USA: 3000 to 2015 by road, rail, and ship.
- Sweden: 40 per year by ship.
- UK: 300 per year by rail.
- France: 250 per year by rail.

- Germany: 40 per year by rail.
- Japan: 200 to 2013 by ship.

(For further information, see information page on Storage and Disposal of Radioactive Wastes, since storage and transport often have a lot in common.)

Plutonium oxide

Plutonium is separated during the reprocessing of used fuel. It is normally then made into mixed oxide (MOX) fuel.

Plutonium is transported, following reprocessing, as an oxide powder, since this is its most stable form. It is insoluble in water and only harmful to humans if it enters the lungs.

Plutonium oxide is transported in several different types of sealed package and each can contain several kilograms of material. Criticality is prevented by the design of the package, limitations on the amount of material contained within the package, and on the number of packages carried on a transport vessel. Special physical protection measures apply to plutonium consignments.

A typical consignment consists of one truck carrying one protected shipping container. The container holds a number of packages with a total weight varying from 80 to 200 kg of plutonium oxide.

Vitrified waste

The highly radioactive wastes (especially fission products) created in the nuclear reactor are segregated and recovered during reprocessing. These wastes are incorporated in a glass matrix by a process known as 'vitrification', which stabilizes the radioactive material. The molten glass is poured into a stainless steel canister where it cools and solidifies. A lid is welded into place to seal the canister. The canisters are then placed inside a Type B cask, similar to those used for the transport of used fuel.

The quantity per shipment depends upon the capacity of the transport cask. Typically a vitrified waste transport cask contains up to 28 canisters of glass.

Return nuclear waste shipments from Europe to Japan since 1995 have been in vitrified form and contained within stainless steel

canisters. Up to 28 canisters (total 14 tonnes) are packed in each 94-tonne steel transport cask, the same type used for the transport of spent fuel. Over 1995-2007 twelve shipments were made from France of vitrified HLW comprising 1310 canisters in total and containing almost 700 tonnes of glass. Return shipments from the UK commenced in 2010 to move about 900 canisters.

Waste materials

Low-level and intermediate-level wastes (LLW and ILW) are generated throughout the nuclear fuel cycle and from the production of radioisotopes used in medicine, industry, and other areas. The transport of these wastes to waste treatment facilities and storage sites is common.

LLW comprises a variety of materials that emit low levels of radiation slightly above normal background levels. It often consists of solid materials, such as clothing, tools, or contaminated soil. LLW is transported from its origin to waste treatment sites, or to an intermediate or final storage facility.

A variety of radionuclides give LLW its radioactive character. However, the radiation levels from these materials are very low and the packaging used for the transport of LLW does not require special shielding.

LLW is transported in drums, often after being compacted in order to reduce the total volume of waste. The drums commonly used contain up to 200 litres of material. LLW is moved by road, rail, and internationally by sea. However, most LLW is only transported within the country where it is produced, given the relative ease of storage and disposal of it.

The composition of ILW is broad, but unlike LLW it requires shielding due to radioactivity levels. Much ILW comes from nuclear power plants and reprocessing facilities.

ILW is taken from its source to an interim storage site, a final storage site (as is the case in Sweden), or a waste treatment facility. It is transported by road, rail, and sea.

In the USA the deep geological repository near Carlsbad, New Mexico, received its 10,000 road shipment of defence-related transuranic waste for permanent disposal in 2011. Almost half the shipments were from Idaho National Laboratory. The repository, known as the Waste Isolation Pilot Plant (WIPP), is about 700 m deep in a Permian salt formation.

Packaging of materials

When radioactive materials are transported, it is important to ensure that radiation exposure to the personnel involved in the transport and the general public along the transport routes is avoided. Packaging for radioactive materials includes, where appropriate, shielding to reduce potential radiation exposure. For some materials, such as fresh uranium fuel assemblies, the radiation levels are negligible and no shielding is required. Other materials, such as used fuel and HLW, are highly radioactive and purpose-designed containers with integral shielding are used. To limit the risk in handling of highly radioactive materials, dual-purpose containers (casks), which are appropriate for both storage and transport of used nuclear fuel, are often used. (For further information, see information page on Storage and Disposal of Radioactive Wastes.)

Arrival of Mox shipment from France to Japan (Image: Kansai Electric Power Company)

As with other hazardous materials being transported, packages of radioactive materials are labelled in accordance with the requirements of national and international regulations. These labels

not only indicate that the material is radioactive by including a radiation symbol, but also give an indication of the radiation field in the vicinity of the package.

The principal assurance of safety in the transport of nuclear materials is the design of the packaging, which must allow for foreseeable accidents. The consignor bears primary responsibility for this as well as for the training of personnel directly involved in the transport. Many different radioactive materials are transported and the degree of potential hazard from these materials varies considerably. Conditions which packages are tested to withstand include: fire, impact, wetting, pressure, heat, and cold. Packages of radioactive material are checked prior to shipping and, when it is found to be necessary, cleaned to remove contamination.

Different packaging standards have been developed by the IAEA according to the characteristics and potential hazard posed by the different types of nuclear material. The IAEA's guidelines are complex, but identify five different categories of primary package based on the activity and physical form of the waste being transported. The categories are: Excepted, Industrial, Type A, Type B, and Type C.

Excepted

Excepted packages have radioactive content at such low levels that the potential hazards are insignificant and therefore no testing is required with regard to containment or shielding integrity.

Industrial

Ordinary industrial containers are used for low-activity material such as uranium oxide concentrate shipped from mines. They are also used for LLW transport within countries.

Type A

'Type A' packages are used for the transport of relatively small, but significant, quantities of radioactive material. They are designed to withstand accidents and are used for limited quantities of medium-activity materials, such as medical or industrial radioisotopes as well as some nuclear fuel materials.

Type B

Type B packages used for HLW, used fuel, and MOX fuel are robust and very secure casks. They range from drum-size to truck-size and maintain shielding from gamma and neutron radiation, even under extreme accident conditions. Designs are certified by national authorities. There are over 150 certified Type B packages, and the larger ones cost around $1.6 million each.

Type B casks for used fuel can weigh up to 110 tonnes when empty and hold from 6 to 20 tonnes of fuel, containing their highly radioactive payload safely during transport. The internal structure of transport casks (using multi-purpose canisters or not) is designed to maintain separation of fuel assemblies even in extreme accidents, and the external structure is designed to maintain safe containment in extreme accidents. Both features are tested before licensing.

An example of a Type B shipping package for used fuel is Holtec's HI-STAR 80 cask (STAR = storage, transport and repository), a multi-layered steel cylinder that holds 12 PWR or 32 BWR high burn-up used fuel assemblies (above 45 GWd/t), and which has had cooling times as short as 18 months.

Holtec's HI-STAR 100 is a high-capacity system which is engineered to accept one sealed multi-purpose canister (MPC) containing a 68-cell fuel basket for BWR fuel or a 32-cell fuel basket for PWR fuel. The MPC containing the fuel can be transferred to HI-STORM 100 storage systems on an independent spent fuel storage installation (ISFSI) pad or below ground surface, exchanging one overpack for another which is engineered for maximum shielding. The HI-STAR 190 cask has a 38 kW heat load capacity and is envisaged as the main used fuel transport to central storage or disposal sites in the USA. It is promoted as a universal transport cask.

Areva has a range of Type B transport casks for used fuel. Its TN12/2 cask designed for used fuel has been adapted for fresh MOX fuel, and holds 12 PWR fuel assemblies or 32 BWR ones. It is robust, with shock-absorbing covers at each end.

In France alone, there are some 750 shipments each year of Type B packages. This is in relation to 15 million shipments classified as 'dangerous goods', 300,000 of which are of radioactive materials of some kind.

In Russia, TUK (transportation packaging set) casks are used to transport used nuclear fuel. Several TUK-13 casks fit into a container or TK carrier for rail transport, each cask holding about 6 tonnes of fuel. A TUK-1410 cask has now been licensed to replace the older model for VVER-1000 fuel, both in Russia and from overseas. Each weighs over 100 tonnes, holds 18 VVER fuel assemblies weighing 9 tonnes in a removable canister, and is designed for hotter fuel – up to 36 kW heat load. (For further information see information page on Storage and Disposal of Radioactive Waste.)

A particular 'Type B' package is used for shipping uranium hexafluoride (UF_6), where the main accident hazard is chemical rather than radiological. Natural uranium is usually shipped to enrichment plants in Type 48Y cylinders, 122 cm diameter and each holding about 12.5 tonnes of uranium hexafluoride. These cylinders are then used for long-term storage of depleted uranium as hexafluoride, typically at the enrichment site. Due to criticality considerations, enriched uranium is shipped to fuel fabricators in smaller Type 30B cylinders, 76 cm diameter and 2.1 m long, each holding 2.27 t UF_6. These may be shipped with overpacks. Both kinds of uranium hexafluoride cylinder must withstand a pressure test of at least 1.4 MPa, a drop test, and survive a fire of 800°C for 30 minutes.

Type C

Smaller amounts of high-activity materials (including plutonium) transported by aircraft are in 'Type C' packages, which give even greater protection than Type B packages in accident scenarios. They can survive being dropped from an aircraft at cruising altitude.

Although not required by transport regulations, the nuclear industry chooses to undertake some shipments of nuclear material using dedicated, purpose-built transport vehicles or vessels.

Purpose-built transportation

In 1993, the International Maritime Organization (IMO) introduced the voluntary *Code for the Safe Carriage of Irradiated Nuclear Fuel, Plutonium and High-Level Radioactive Wastes in Flasks on Board Ships* (INF Code), complementing the IAEA Regulations. These provisions mainly cover ship design, construction, and equipment. The INF Code came into force in January 2001 and introduced

advanced safety features for ships carrying used fuel, MOX, or vitrified HLW.

There are at least five small purpose-built ships ranging from 1250 to 2200 DWT, and four purpose-built ships ranging from 3800 to 5000 DWT, able to carry Type B casks and other materials. They conform to all relevant international safety standards, notably INF-3 (Irradiated Nuclear Fuel class 3) set by the IMO, allowing them to carry highly radioactive materials such as HLW and used nuclear fuel, as well as mixed-oxide (MOX) fuel, and plutonium.

The three largest ships belong to a British-based company, Pacific Nuclear Transport Ltd (PNTL), a subsidiary of International Nuclear Services Ltd (INS)*. The three PNTL vessels currently in service, the *Pacific Heron*, *Pacific Egret*, and *Pacific Grebe* were launched in Japan in 2008, 2010, and 2010, respectively. All have double hulls separated by impact-resistant structures, together with duplication and separation of all essential systems to provide high reliability and significant contingency in the event of an accident. Twin engines operate independently. Each ship can carry up to 20 or 24 transport casks. Each ship is 4916 tonnes (DWT) and 104 metres long. *Pacific Grebe* carries mainly wastes, whilst the other two usually carry consignments of MOX fuel. Earlier ships in the PNTL fleet mainly carried Japanese used fuel to Europe for reprocessing. The PNTL fleet has completed more than 180 shipments with more than 2000 casks over some 40 years without any incident resulting in the release of radioactivity.[9]

* PNTL is now owned by International Nuclear Services Ltd (INS, 68.75%), Japanese utilities (18.75%), and Areva (12.5%). INS is in turn owned by the UK's Nuclear Decommissioning Authority.

In 2013 Sweden's SKB commissioned the *Sigrid*, a slightly larger replacement for its 1982 vessel the *Sigyn*. *Sigrid* was built by Damen Shipyards and carried its first shipment in January 2014. It is used for moving used fuel from reactors to the CLAB interim waste storage facility. *Sigrid* is equipped with a double hull, four engines, and redundant systems for safety and security. *Sigrid* is 99.5 metres long and 18.6 metres wide, 1600 DWT and capable of carrying 12 nuclear waste casks. (*Sigyn* was 1250 DWT and carried ten casks.)

Rosatomflot is operating the 1620 DWT *Rossita*, built in Italy and completed in 2011. It is designed for transporting spent nuclear fuel and materials of decommissioned nuclear submarines from Russian Navy bases in northwest Russia. It will be used on the Northern Sea Route, between Gremikha, Andreeva Bay, Saida Bay,

Severodvinsk, and other places hosting facilities which dismantle nuclear submarines. Spent fuel is to be delivered to Murmansk for rail shipment to Mayak. Rosatomflot has the *Serebryanka* (1625 DWT, 102 metres long, built 1974) already in service. The *Imandra* (2186 DWT, 130 metres long, built 1980) is described as a floating technical base but is reported to be in service transporting used fuel and wastes from the Nerpa shipyard and Gremikha to Murmansk.

Rossita is an ice-class vessel and is designed to operate in the harsh conditions of the Arctic. The ship is 84 metres long and 14 m wide, has two engines, and two isolated cargo holds with a total capacity of 720 tonnes. The €70 million vessel was given to Russia as part of Italy's commitment to the G8 partnership program for cleaning up naval nuclear wastes, and is designed to cover all needs in spent nuclear fuel and radwaste shipments in northwest Russia throughout the entire period of cleaning up these territories.

Rosatomflot also operates a new vessel, the semi-submersible pontoon dock *Itarus*, built in Italy under a 2013 contract, and delivered in 2016. It is designed to transport three compartment units of dismantled Russian nuclear submarines for SevRAO in Saida Bay.

In January 2021 China National Nuclear Corporation (CNNC) announced that it had taken delivery of its first ship designed for transport of used nuclear fuel. The *Xin An Ji Xiang* meets INF-3 standards set by the IMO.

Road transport of used fuel, Japan (Image: Nuclear Fuel Transport Ltd.)

Accident scenarios

There has never been any accident in which a Type B transport cask containing radioactive materials has been breached or has leaked. A significant accident in the USA in 1971 demonstrated the integrity of a Type B cask, which was later returned to service.

The safety features built into Type B containers are very significant. For the radioactive material in a large Type B package in sea transit to become exposed, the ship's hold (inside double hulls) would need to rupture, the 25 cm thick steel cask would need to rupture, and the stainless steel flask or the fuel rods would need to be broken open. Either borosilicate glass (for reprocessed wastes) or ceramic fuel material would then be exposed, but in either case these materials are very insoluble.

The purpose-built transport ships described above are designed to withstand a side-on collision with a large oil tanker. If the ship did sink, the casks would remain sound for many years and would be relatively easy to recover since instrumentation including location beacons would activate and monitor the casks.

Challenges in radioactive material transport

The majority – over 95% – of radioactive material consignments relate to radioisotopes for medical and industrial use (for further information see information paper on The Many Uses of Nuclear Technology). A 2015 Euratom Supply Agency study identified lack of harmonization and over-regulation in transport authorization for radioactive materials, particularly between countries, as a significant risk from a security of supply perspective.

Multiple layers of regulation and a lack of international consistency are considerable disincentives, and may deter companies from executing shipments. Shipments are occasionally denied due to national competent authorities not being recognized by other countries.

Most reports of denial of shipment relate to non-fissile materials, either Type B packages (mainly cobalt-60) or tantalum-niobium concentrates. For uranium concentrates the main problem is the limited number of ports which handle them, and the relatively few marine carriers which accept them. For all radioactive materials, consignors are required to provide training to personnel handling the packages, creating significant cost and inconvenience to shippers.

Notes

Any goods that pose a risk to people, property and the environment are classified as dangerous goods, which range from paints, solvents and pesticides up to explosives, flammables, and fuming acids, and are assigned to different classes under the UN Recommendations on the Transport of Dangerous Goods, Model Regulations:

- Class 1: Explosives.
- Class 2: Gases.
- Class 3: Flammable liquids.
- Class 4: Flammable solids; substances liable to spontaneous combustion; substances which, on contact with water, emit flammable gases.
- Class 5: Oxidizing substances and organic peroxides.
- Class 6: Toxic and infectious substances.
- Class 7: Radioactive material.
- Class 8: Corrosive substances.
- Class 9: Miscellaneous dangerous substances and articles, including environmentally hazardous substances.

When transported, these goods need to be packaged correctly, as laid out in the various international and national regulations for each mode of transport, to ensure that they are carried safely to minimize the risk of an incident.

The US Nuclear Regulatory Commission defines, for transport purposes only, radioactive materials as those with specific activity greater than 74 Bq per gram. This definition does not specify a quantity, only a concentration. As an example, pure cobalt-60 has a specific activity of 37 TBq per gram, which is about 500 billion times greater than the definition. However, uranium-238 has a specific activity of only 11 kBq per gram, which is only 150 times greater than the definition.

Sources

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